Rayapati Pushkar

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.84
Narayan Junior College	Class XII (IPE)	2024	98.60%
Narayana E.M High School	Class X (SSC)	2022	92.34%

SCHOLASTIC ACHIEVEMENTS

- **Indian Olympiad Qualifier in Mathematics (IOQM) 2022** - Qualified, One of **629** students nationwide selected for the next stage.
- **AlgoUniversity Technology Fellowship (ATF) 2025** - Qualified Stage 1, ranked in the **top 8% nationwide** among **250,000 participants**; advanced to Stage 2.

RELEVANT COURSES

- **Mathematics:** Probability, Applied Statistics, Linear Algebra, Convex Optimization, Transform Techniques, Intro to Metric Spaces, Number Theory, Ordinary Differential Equations, Calculus I and II.
- **Computer Science:** Introduction to Programming, Data Structures and Applications, Algorithms, Database Management Systems I, Theory of Computation, Compilers I

SKILLS

- **Tools & Platforms:** Linux(Ubuntu), VS Code, GitHub.
- **Programming:** C, C++, Python, JavaScript, SQL.
- **Web Technologies:** HTML, CSS, Django.

PROJECTS

- **RouteX - Smart Delivery Routing System** | *Python, Django, HTML, CSS, JavaScript*(View Project)
- Built a full-stack **Django logistics platform** supporting role-based authentication, order lifecycle management, and delivery partner assignment.
- Implemented **Dijkstra's shortest-path algorithm** over a weighted graph to identify optimal delivery routes and assign the nearest available delivery partner.
- Developed an interactive map interface using **Leaflet.js** and **OpenStreetMap**, while structuring the backend with reusable service modules and transactional database operations.
- Applied graph modelling and path reconstruction to optimize delivery assignment while maintaining a modular and extensible backend.
- Designed a modular backend by separating routing algorithms, business logic, and presentation layers to improve maintainability and extensibility.

- **LZ77 Compression Engine using Suffix Tree** | *C++, STL*(View Project)
- Implemented the **LZ77 lossless text compression algorithm** in C++ using a suffix-tree-based data structure for efficient longest-match search.
- Designed a modular architecture with separate components for compression logic, suffix tree construction, and file processing.
- Encoded input text into **(offset, length, next character)** tokens while supporting configurable search and look-ahead windows.
- Applied string processing and custom data structures without relying on external compression libraries.

- **IndusMind AI** | Team Hackathon Project | *Django, Python, JavaScript, LangChain, ChromaDB*(View Project)
- Developed responsive frontend interfaces for an enterprise industrial intelligence platform using Django Templates, HTML, CSS and JavaScript.
- Built dashboards for AI Copilot, document management, AI insights and knowledge graph visualization.
- Collaborated on integrating frontend components with a **Retrieval-Augmented Generation (RAG)** pipeline, knowledge graph and AI agent architecture.
- Designed reusable UI components and **REST API** integration for document intelligence workflows.

POSITIONS OF RESPONSIBILITY

- **Operations Domain Coordinator, EBSB** - Ek Bharat Shrestha Bharat (FY 2025-26)
- Collaborated with fellow coordinators and volunteers to manage event logistics and ensure smooth execution.

EXTRACURRICULAR

- Participated in the **Economic Times Hackathon 2026**, collaborating in a team to develop an enterprise AI solution using Django, LangChain and ChromaDB.

- As part of **National Service Scheme(NSS)**, participated in community awareness campaigns and walkathons promoting health, social responsibility and public welfare.